4 Control Valves

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Topic Highlights

Valve Types
Standards and Codes
Valve Selection
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4.1 Introduction

A control valve is a power-actuated device that modifies the fluid flow rate in a process control system. The valve is connected to an actuator mechanism that is capable of changing the position of the valve's closure member in response to a signal from the controlling system.

A control valve is used to control properties such as upstream pressure, downstream pressure, flow rate, or liquid level. The control valve responds to a controller that measures and compares one of these properties to a setpoint. The controller's signal to the control valve will vary to maintain the property at the setpoint. Functionally, a control valve can be described as a controlled variable orifice.

The control valve's actuator positions the closure member, in the valve body, in a position consistent with the controller's signal. The closure member, often called a plug or a disc, creates a flow area in conjunction with a seat or a guiding cage that modifies the flow rate.

4.2 Valve Types

Control valves have two basic styles: linear and rotary motion. The valve stem of linear motion valves moves linearly up and down. The valve shaft of rotary motion valves rotates without any linear motion. A globe valve is a typical linear motion valve; ball valves and butterfly valves are both rotary motion valves. Linear motion valves are commonly used for more severe duty, with the rotary motion valves generally used in moderate to light duty service in sizes above 2-inch. For the same line size, rotary valves are smaller and lighter than linear motion valves and the rotary motion valves are more economical in cost, particularly as the line sizes increase.

A ball valve, used as a control valve, will usually have design modifications to improve its performance. Instead of a full spherical ball, it will typically have a 1/3 ball segment. This reduces the amount of seal contact and, therefore, reduces friction, allowing for more precise positioning. The leading edge of the ball segment may have a V-shaped groove to improve the control characteristic. A ball valve's trim material is generally 300 series stainless steel.

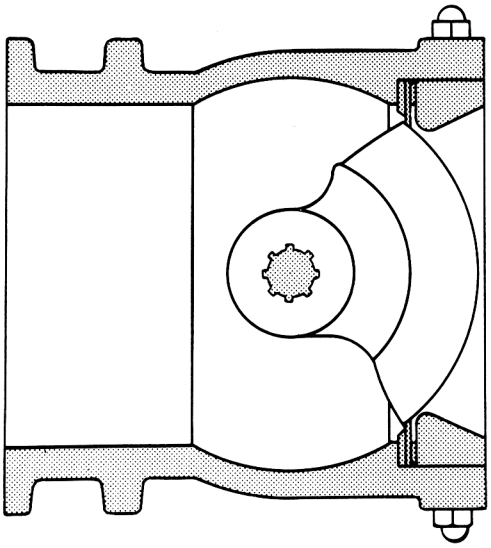
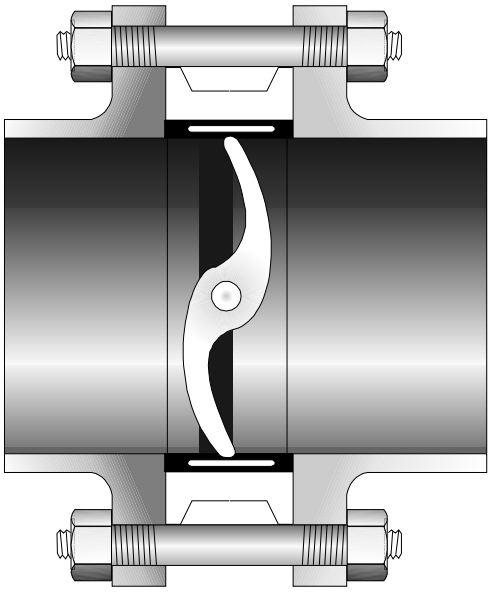


Figure 4-1: Segmental Ball Valve Cross Section (Courtesy: Masoneilan/Dresser)

A butterfly valve used as a control valve may have a somewhat S-shaped disc to reduce the flow-induced torque on the disc, allowing for more precise positioning. A butterfly valve's trim material may be bronze, ductile iron, or 300 series stainless steel.



LINE FLANGES ANCHOR PUSH-IN LINER.
SOURCE: MASONEILAN/DRESSER

Figure 4-2: Butterfly Valve (Courtesy: Masoneilan/Dresser)

Another rotary control valve is the eccentric disc with the closure member shaped similar to a mushroom and is attached slightly offset to the shaft. The style provides good control along with tight shutoff, as the offset supplies leverage to flex the disc face into the seat. This valve's advantage is tight shutoff without the elastomeric seat seals that are used in ball and butterfly valves. Eccentric disc valves' trim material is generally 300 series stainless steel, which may be clad with stellite hardfacing.

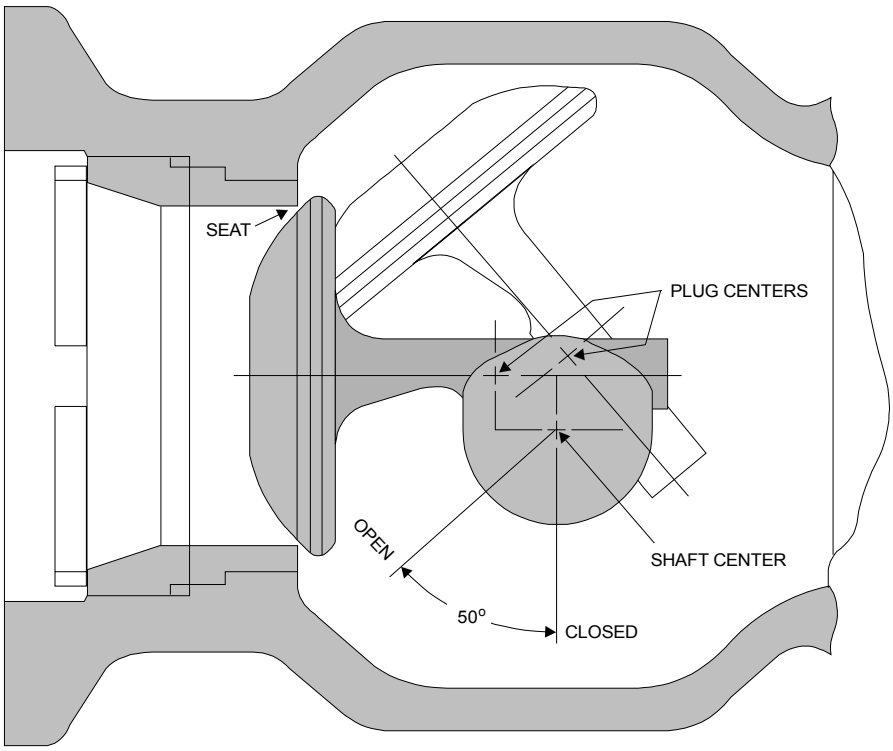


Figure 4-3: Eccentric Control Valve

Linear motion control valves have two common styles: post-guided and cage-guided.

Post-guided valves have the moving closure member guided by a bushing in the valve's bonnet. The closure member is usually unbalanced, and the fluid pressure drop acting on the closure member can create significant forces. Post-guided trims are well suited for slurries and fluids with entrained solids. The post-guiding area is not in the active flow stream, reducing the chance of solids entering the guiding joint. Post-guided valve trim materials are usually either 400 or 300 series or 17-4PH stainless steel.

Cage-guided valves have a cylindrical cage between the body and bonnet. Below the cage is a seat ring. The cage/seat ring stack is sealed with resilient gaskets on both ends. The cage guides the closure member, also known as the plug. The plug is often pressure balanced with a dynamic seal between the plug and cage. The balanced plug will have passageways through the plug to eliminate the pressure differential across the plug and the resulting pressure induced force. The trim materials for cage guided valves are often either 400 or 300 series or 17-4PH stainless steel sometimes nitrided or stellite hard faced.

The valve closure member is connected to the actuator with either a stem or a shaft. Stem is the term used for linear motion valves, and shaft is used for rotary motion. The stem/shaft connects from inside the valve body to the actuator outside. The fluid pressure is sealed with a packing box; a cylindrical

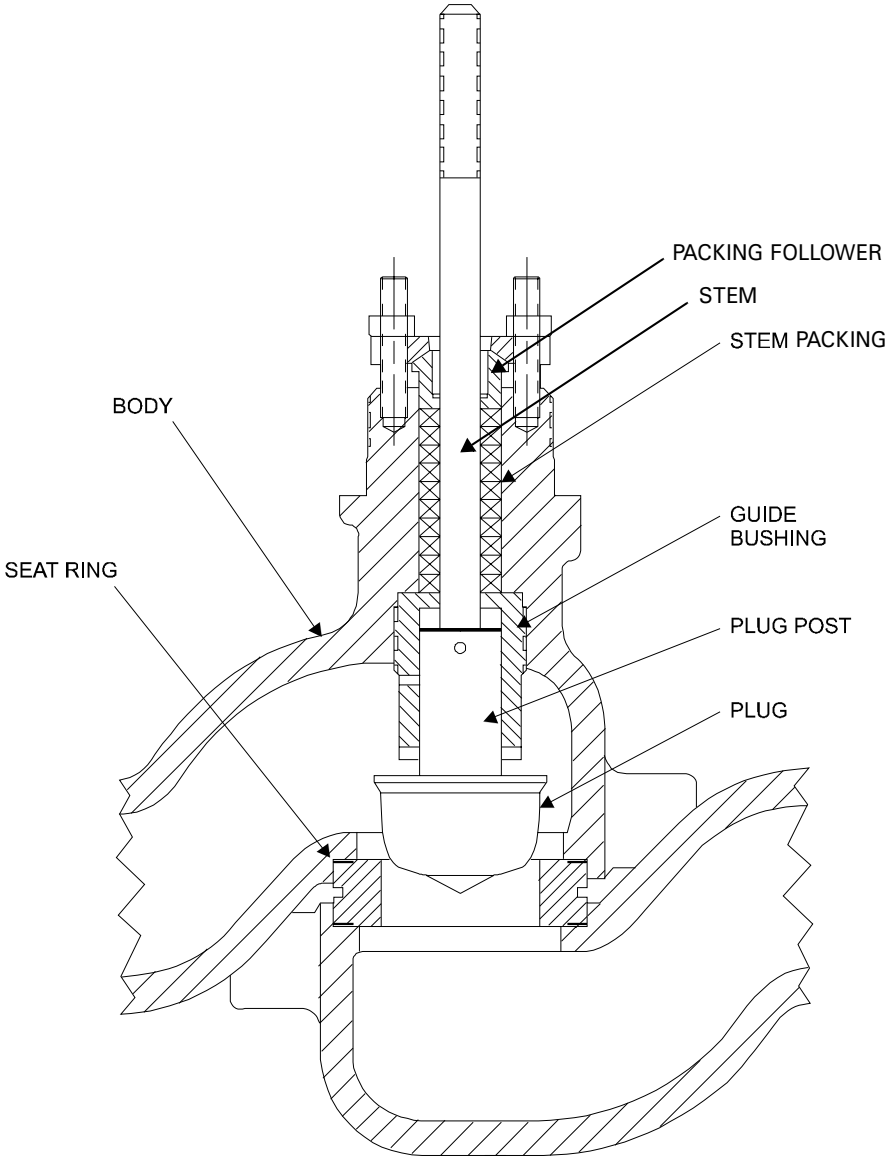


Figure 4-4: Post-Guided Control Valve

chamber with a guiding bushing; generally a packing spring or other means of applying a resilient compressive force to the packing; several rings of packing; and a packing follower. The packing rings need to be compressed to effect a tight seal. The fluid pressure will usually provide sufficient compression force on the packing rings. The packing rings usually lose some of their volume, reducing the stacked height of the packing. The packing spring serves to accommodate the volume change of the packing rings as well as providing a sufficient initial force as the valve is being pressurized to ensure a tight seal.

4.3 Standards and Codes

A control valve is a pressure vessel and must meet an industry code such as ASME/ANSI B16.34 - 1996 - *Valves Flanged, Threaded, and Welding End*. This code provides calculations for minimum wall thicknesses, along with temperature/pressure ratings for 40 different body materials. It also provides rules for marking and hydrostatic testing. Valves have pressure ratings indicating their allowed working pressures at various temperatures. The pressure ratings are Classes 150, 300, 400, 600, 900, 1500,

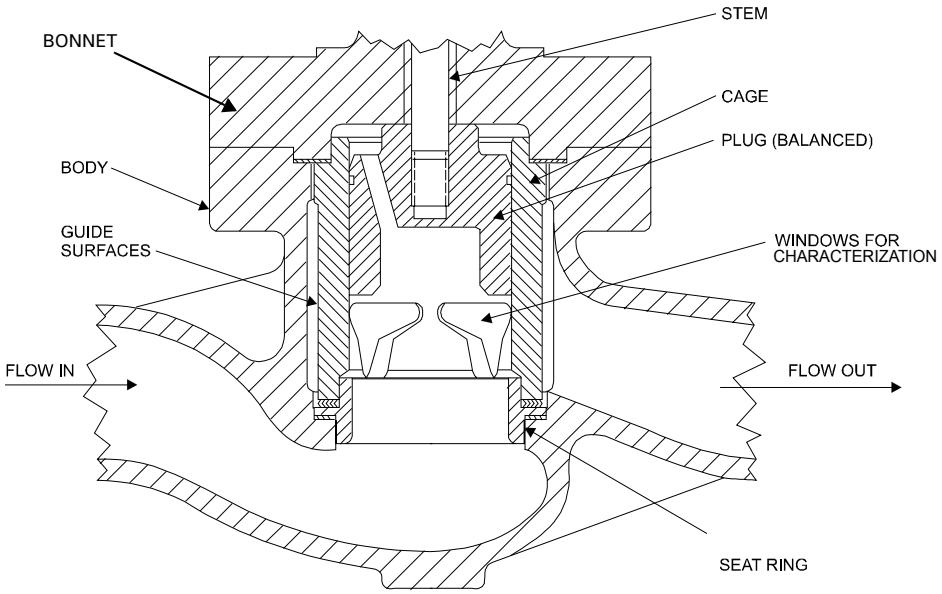


Figure 4-5: Cage-Guided Control Valve

2500, and 4500. The pressure class does not directly relate to a specific pressure/temperature relationship but is a generic designation for the pressure rating class. The class number matches the allowable pressure at an intermediate temperature around 600°F to 800°F. The valve's body and bonnet materials are, most commonly, carbon steel, chrome moly steel, and stainless steel. Chrome moly steel is used for elevated temperature applications such as power generation, and stainless steel is used for corrosive applications such as petrochemical.

There is a broad range of valve end connections used to connect a control valve to a pipeline: flanged, butt weld, socket weld, threaded end, and flangeless. The flangeless connection is installed between two pipeline flanges using long studs to clamp the valve between the flanges.

4.4 Valve Selection

Choosing the appropriate control valve for a particular application is a multistep process. The initial and most important consideration is determining the required flow capacity by calculating the valve sizing coefficient, C_V. The basic formula for incompressible fluids is

$$C_V = Q\sqrt{\frac{G_f}{\Delta P}}$$

where

Q is the flow rate

G_f is the fluid's specific gravity

ΔP is the pressure drop across the valve

ANSI/ISA-75.01.01-2002 (60534-2-1 Mod) - *Flow Equations for Sizing Control Valves* fully describes the variations of the basic formula for liquid, vapor and two phase flows with consideration for cavitation, flashing, and sonic velocity—all affecting the required C_V. Most control valve manufacturers have flow

sizing programs available for use on their Web sites. Manufacturers rate their valves to the maximum and minimum controllable C_V to enable the proper selection of valve for the particular application.

Control valves create a pressure drop in the flow stream to modify the flow rate. A reduced flow area is developed by the position of the closure member relative to the seat or cage port. The flowing fluid loses pressure and accelerates as it passes through the reduced flow area. The point of highest fluid velocity and lowest pressure is called the vena contracta. Downstream of the vena contracta, the fluid pressure partially recovers. The degree of pressure recovery is a major element in choosing the appropriate control valve for a particular situation.

Figure 4-6 shows the pressure drops and pressure recovery as fluid passes through a restriction. In the example shown, the lowest pressure is at the vena contracta (P_{VC}) with the pressure recovering about 35% to the outlet pressure. The vena contracta pressure becomes very significant for liquid flow when the fluid vapor pressure is greater than the vena contracta pressure. If the liquid vapor pressure shown as Vapor Pressure 1 is higher than the vena contracta pressure and below the outlet pressure, then liquid vaporizing at the vena contracta will recondense to a liquid, resulting in cavitation in the region between Vapor Pressure 1 and the vena contracta. If the vapor pressure shown as Vapor Pressure 2 is higher than the outlet pressure, then the liquid will vaporize or "flash" leaving the valve as a liquid/vapor mixture. Each of these phenomena can cause operational problems.

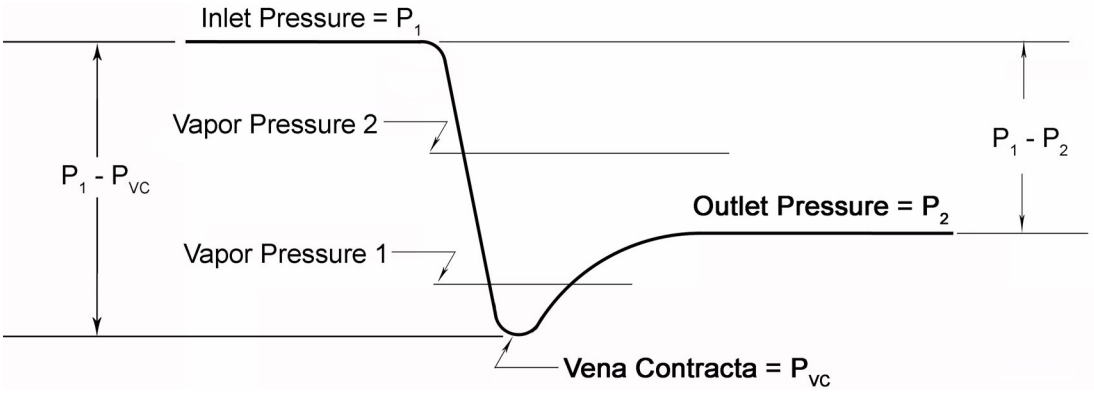


Figure 4-6: Pressure Drops and Recovery as Fluid Passes Through Restriction

Figure 4-6 shows the flow profile for a typical linear motion globe valve. A rotary motion valve, such as a ball or butterfly valve, with the same inlet and outlet pressures will require the pressure drop between the inlet pressure and the vena contracta to be about 75% larger, for the same overall pressure drop. This is due to its straight-through, streamlined flow path and higher pressure recovery characteristic. This phenomenon makes these valves more susceptible to cavitation.

The vena contracta pressure is also significant for gas and vapor flow. The gas or vapor flow rate cannot exceed the sonic velocity. The valve's maximum flow velocity occurs at the vena contracta and, once the flow velocity becomes sonic, the flow is choked and further reductions in the downstream pressure will not increase the flow rate through the valve.

Cavitation occurs in liquid flow when the fluid pressure drops below the liquid's vapor pressure and the vapor pressure is below the outlet pressure. In this region, the liquid will flash to a vapor and then suddenly collapse back to a liquid. This sudden collapse creates very high localized shock waves that will cause surface damage when it occurs at, or adjacent to, components of the valve.

Flashing occurs in liquid flow when the fluid pressure drops below the liquid's vapor pressure and the vapor pressure is above the outlet pressure. The liquid will continue to flash downstream of the valve

as the fluid pressure decreases. The flowing mixture of liquid and vapor may cause erosion of the valve and pipe surfaces and create a higher flow noise level than for liquid flow.

Cavitation and flashing flow with entrained vapor may limit the flow rate due to the increased volume of the liquid/vapor.

4.5 Operation

Effective control valve operation requires a rigorous application process. The operating conditions must be known, as most misapplications come from having incorrect or incomplete information. The application information may include identification of the flowing fluid, along with its viscosity and density, inlet and outlet pressures, fluid temperature, and range of flow rates. Not all applications require each of these data elements, but the more severe service requires a comprehensive specification. The primary element is the calculation of the C_V value.

The next step is usually choosing between linear or rotary motion valves. If there is a wide range in the flow rate, the minimum C_V should be calculated. Rangeability is the ratio of maximum to minimum controllable C_V values. Each valve style has its rangeability limits. Fluid flow noise is also a consideration, since excessive flow noise, above the permissible noise levels of OSHA regulations, can harm a person's hearing permanently and can shorten valve life. Aerodynamic noise from vapor flow can be predicted using the standard ISA-75.17-1989 - *Control Valve Aerodynamic Noise Prediction*. The C_V calculation process will indicate if cavitation or flashing is likely and also if the flow rate will be choked from cavitation, flashing, or sonic velocity.

Control valves are seldom fully open or closed and generally are moving in response to the control signal. However, when the valves are closed, the allowable seat leakage is important, affecting the choice of trim style, seat material, and actuator size.

A control valve's seat tightness is specified by Leakage Classes as defined in the ANSI/FCI 70-2 *Control Valve Leakage* standard. Six leakage classes are specified: Classes I, II, III IV, V, and VI. Classes I through VI progressively require increasingly stringent sealing performance. As examples, Class II allows 0.5% of the valve's rated flow capacity (that is .5% of the valve's rated maximum C_V). Class VI, normally applied to resilient seated valves, allows 0.45 ml of liquid leakage per minute at 50 psi for a 2" valve.

The required C_V value can be calculated using ANSI/ISA-75.01.01-2002 (60534-2-1 Mod) - *Flow Equations for Sizing Control Valves*. Aerodynamic noise from vapor flow can be predicted using ISA-75.17-1989 - *Control Valve Aerodynamic Noise Prediction.*

The relationship between the valve's trim travel and the trim's flow coefficient, C_V, is the "Inherent Characteristic" determined by flow testing with a constant differential pressure applied across the valve throughout the valve travel. The two more common characteristics are Equal Percentage and Linear.

The linear characteristic C_V value is linear to the valve travel. The equal percentage characteristic is defined as an equal percentage increase in C_V for each equal increment of valve travel. The equal percentage C_V value increases slowly at the lower end of the valve travel and then increases more rapidly as the trim opens. Inherent valve flow characteristics are chosen with respect to operating conditions such that the resulting installed characteristic is approximately linear. An equal percentage characteristic has two important advantages: it offers more precise control at the lower end of travel and, in applications where the pressure drop across the valve significantly decreases as the valve opens, the resulting installed characteristic approximates a linear characteristic, providing effective control.

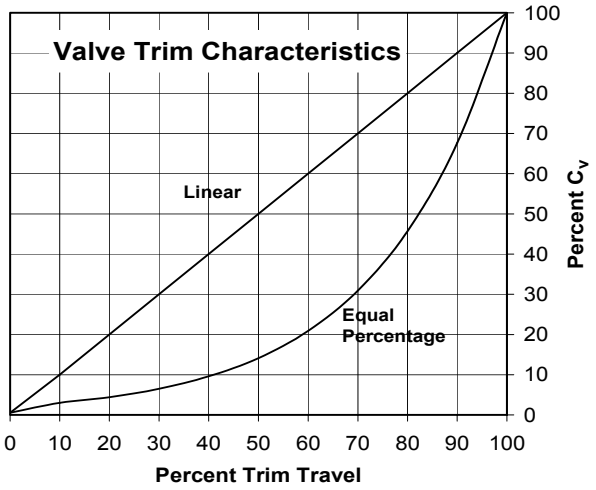


Figure 4-7: Valve Trim Characteristics

4.6 Actuators and Accessories

A control valve actuator will precisely position the closure member in response to the control signal. There are three major groups of actuators: pneumatic, electric, and hydraulic.

The most common is the pneumatic actuator, mainly in a spring/diaphragm style but also in a piston style, with or without a spring. The spring in a pneumatic actuator opposes the force of the pneumatic pressure—usually air—against the diaphragm or piston. The pneumatic pressure is increased to create motion by compressing the spring, or the pressure is reduced to allow motion in the opposite direction by partially uncompressing the spring. The spring/diaphragm actuator can be used for rotary motion valves by converting the linear motion to rotary with lever linkage. The spring/diaphragm actuator can be configured so the spring can open the valve (direct acting) or so the spring can close the valve (reverse acting). The reverse acting mode will fail closed when actuating pressure is lost. The direct acting mode will fail open.

Pneumatic actuators' main advantage is lower cost than the other styles, together with low friction and low dead band.

The hydraulic actuator is very similar to the pneumatic actuator, except its force is provided by pressurized liquid. A hydraulic actuator's style is usually piston. An advantage of a hydraulic actuator is the actuating fluid is incompressible, resulting in a more stable device that is less affected by rapidly changing closure member forces than the pneumatic actuator. The hydraulic actuator's other advantage is its construction allows higher actuating pressures.

The electric actuator uses an electric motor with a gear train to rotate a threaded stem nut, for linear valves, or a shaft for rotary valves.

Actuators have three major accessories: positioners, limit switches, and pressure transducers.

A positioner measures the valve's stem/shaft position, comparing it to the signal. The positioner will adjust the force applied by the actuator—whether pneumatic, hydraulic, or electric—to maintain the position of the valve closure member relative to the signal. The positioner's signal may be pneumatic, electric (variable current), or a digital electronic input. A positioner can be considered a valve controller. A digital "smart" positioner can communicate back to the controller, sending the positioner's actual position, supply pressure, actuator pressure, and a performance diagnosis of itself and the valve/actuator system.

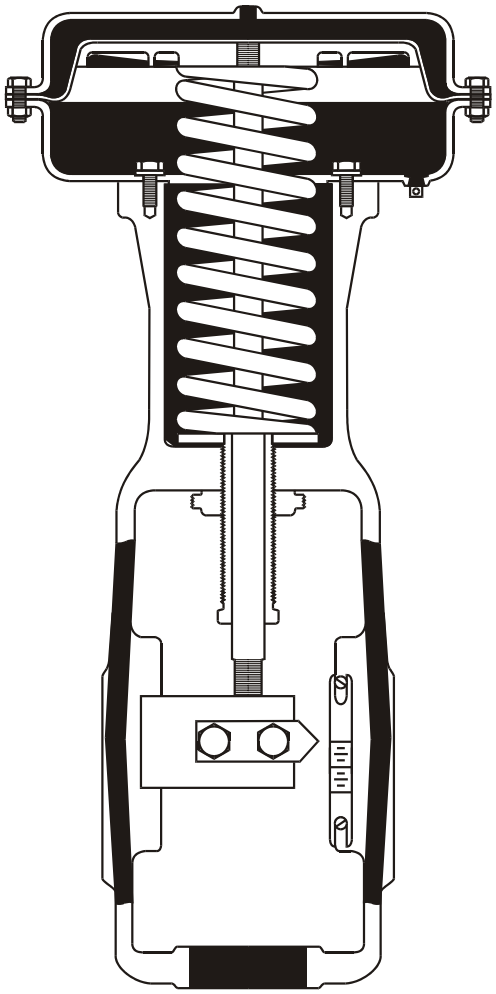


Figure 4-8: Direct Acting Spring/Diaphragm Actuator (Courtesy: Norriseal)

The limit switches monitor the stem/shaft position, opening or closing the switch's contacts when the stem/shaft is in a particular position, usually fully open or closed.

Pressure transducers are usually used to convert an electric input signal to pneumatic or hydraulic pressure.

Precision positioning is a major goal of a control valve actuator, with friction being a major cause for losses of precision. Friction may also create significant deadband. Deadband is the range the signal may be varied, upon reversal of direction, without stem/shaft motion. If the static friction is much higher than the dynamic friction, the pneumatic actuator will increase pressure to overcome the high static friction and then may surge past its intended position, as the lower dynamic friction cannot contain the excess pressure. After this happens, the positioner reverses direction and tries to achieve the intended position again, overshooting going into a hunting cycle. High friction can also cause hysteresis between the increasing signal position and the position of the same decreasing signal. Some valve applications require very accurate control. Achieving this requires an accurate positioner and low friction on all moving parts.

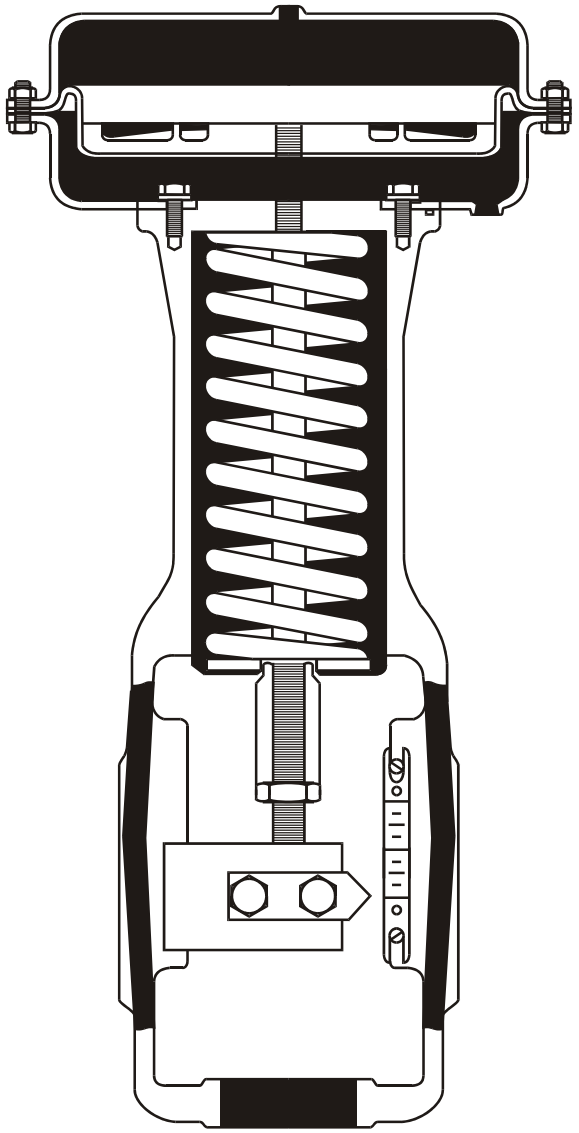


Figure 4-9: Reverse Acting Spring/Diaphragm Actuator (Courtesy: Norriseal)

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About the Author

James Reed, now retired, was Vice President of Engineering for Norriseal in Houston. He is a graduate of Pennsylvania State University. He has been responsible for product design/development and technical personnel management for three control valve companies: Norriseal, Masoneilan, and Copes Vulcan. He has been an active member of seven ISA SP75 standards subcommittees for more than 20 years and is currently chairman of two.